THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE CHAPTER TEST SET A

Chapter 8: **Sound Waves: Characteristics and Applications** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. At a point in a medium, 10 density oscillations occur in 2 s. The frequency of the sound wave is: 1

(a) 5 Hz **(b)** 12 Hz **(c)** 0.2 Hz **(d)** 10 Hz

2. A sound wave of frequency 5 Hz has a time period of: 1

(a) 5 s **(b)** 0.4 s **(c)** 0.1 s **(d)** 0.2 s

3. A source of frequency 680 Hz produces waves of wavelength 0.5 m in a medium. The speed of the wave is: 1

(a) 340 m s^{-1} **(b)** 1360 m s^{-1} **(c)** 680 m s^{-1} **(d)** 680.5 m s^{-1}

4. Sound of frequency 680 Hz travels at 340 m s^{-1}. Its wavelength is: 1

(a) 2 m **(b)** 231200 m **(c)** 1 m **(d)** 0.5 m

5. An echo is heard 0.2 s after a shout. If the speed of sound is 340 m s^{-1}, the distance of the reflecting surface is: 1

(a) 34 m **(b)** 44 m **(c)** 68 m **(d)** 17 m

6. A sonar signal returns from the ocean floor after 5 s. If the speed in water is 1525 m s^{-1}, the depth is: 1

(a) 7625 m **(b)** 1906.25 m **(c)** 3822.5 m **(d)** 3812.5 m

7. Sound cannot travel through: 1

(a) steel **(b)** vacuum **(c)** water **(d)** air

8. Elephants can detect: 1

(a) ultrasound only **(b)** light waves **(c)** infrasound **(d)** radio waves

9. Sound travels fastest in: 1

(a) gases **(b)** solids **(c)** liquids **(d)** vacuum

10. Persistence of sound due to multiple reflections is called: 1

(a) resonance **(b)** pitch **(c)** echo **(d)** reverberation

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. How is sound produced? Give an example. 2

12. Why can astronauts not talk directly to each other on a spacewalk? 2

13. Define compression and rarefaction. 2

14. Why is sound called a longitudinal wave? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Describe an activity to show that sound is produced by vibrations. 3

16. Describe the bell jar experiment and state the conclusion. 3

17. Explain how sound propagates in air using the piston model. 3

OR

Distinguish between longitudinal and transverse waves with one example each. 3

18. Explain with the help of a graph the terms crest, trough and wavelength. 3

OR

If 10 density oscillations occur in 2 s, find the frequency and time period. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) How is sound produced? Describe an activity. 5
(ii) Show that sound needs a medium.
(iii) Why can astronauts not hear each other directly?

OR

(i) Explain the propagation of sound as compressions and rarefactions. 5
(ii) Is the medium carried along?
(iii) Distinguish longitudinal and transverse waves.

20. **Case study:** A tuning fork of frequency 256 Hz is struck; the speed of sound in air is 344 m s^{-1}. 5
(i) Find the time period.
(ii) Find the wavelength.
(iii) How many compressions pass a point in 5 s?
(iv) What happens to the wavelength in water if the speed there is 1500 m s^{-1}?

OR

(i) Define amplitude, frequency, wavelength and time period. 5
(ii) Relate loudness to amplitude and pitch to frequency.
(iii) Give the audible range.

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND